

## **TASK 1:-**

### **Physically assemble a workstation and document the component installation sequence.**

- To physically assemble a computer workstation by installing all hardware components in the correct order and verifying successful assembly.

#### **Components Required**

- Computer Cabinet (Case)
- Motherboard
- Processor (CPU)
- CPU Cooler/Fan
- RAM (Memory Module)
- SMPS (Power Supply Unit)
- SSD/HDD (Storage Device)
- Graphics Card
- SATA Data Cable
- SATA Power Cable
- Keyboard
- Mouse
- Monitor
- Power Cable
- Screwdriver
- Anti-static wrist strap

#### **Installation Sequence**

##### **Step 1: Prepare the**

##### **Workspace**

- Switch off the power supply.
- Wear an anti-static wrist strap.
- Keep all components on a clean, non-conductive surface.

##### **Step 2: Install the Power Supply (SMPS)**

- Place the SMPS inside the cabinet.

- Align it with the mounting holes.
- Secure it using screws.

### **Step 3: Install the CPU on the Motherboard**

- Open the CPU socket lever.
- Match the CPU's alignment mark with the socket.
- Gently place the CPU into the socket.
- Lock the lever.

### **Step 4: Apply Thermal Paste**

- Apply a small amount of thermal paste on the CPU.
- Skip this step if the CPU cooler already has pre-applied paste.

### **Step 5: Install the CPU Cooler**

- Place the cooler over the CPU.
- Lock or screw it into place.
- Connect the cooler cable to the **CPU\_FAN** header.

### **Step 6: Install RAM**

- Open the RAM slot clips.
- Align the notch on the RAM with the slot.
- Press firmly until both clips lock.

### **Step 7: Install the Motherboard**

- Install motherboard standoffs inside the cabinet.
- Place the I/O shield (if separate).
- Position the motherboard carefully.
- Secure it with screws.

### **Step 8: Install Storage Device (SSD/HDD)**

- Mount the SSD or HDD in the drive bay.
- Secure it with screws.
- Connect the SATA data cable to the motherboard.
- Connect the SATA power cable from the SMPS.

### **Step 9: Install Graphics Card (Optional)**

- Insert the graphics card into the PCIe x16 slot.
- Secure it with screws.
- Connect PCIe power cables if required.

### **Step 10: Connect Power Cables**

- Connect the 24-pin ATX motherboard power cable.
- Connect the 4-pin or 8-pin CPU power cable.
- Connect storage power cables.
- Connect GPU power cable (if required).

### **Step 11: Connect Front Panel Connectors**

Connect:

- Power Switch (PWR SW)
- Reset Switch (RESET SW)
- Power LED
- HDD LED
- Front USB
- Front Audio

### **Step 12: Cable Management**

- Arrange cables neatly.
- Use cable ties.
- Ensure no cable blocks cooling fans.

### **Step 13: Connect External Devices**

- Monitor
- Keyboard
- Mouse
- LAN Cable (optional)
- Power Cable

### **Step 14: Power On and Test**

- Turn on the SMPS.
- Press the power button.
- Check if:
  - CPU fan spins.
  - Display appears.
  - BIOS opens successfully.

- RAM and storage are detected

| Component                               | Status |
|-----------------------------------------|--------|
| CPU Installed                           | ✓      |
| CPU Cooler Connected                    | ✓      |
| RAM Installed                           | ✓      |
| Motherboard Mounted                     | ✓      |
| SMPS Installed                          | ✓      |
| SSD/HDD Connected                       | ✓      |
| Component                               | Status |
| Graphics Card Installed (if applicable) | ✓      |
| Power Cables Connected                  | ✓      |
| Front Panel Connected                   | ✓      |
| System Boots Successfully               | ✓      |

## Precautions

- Handle components by their edges.
- Never force components into their slots.
- Prevent electrostatic discharge (ESD).
- Double-check all power connections before turning on the system.
- Ensure proper airflow inside the cabinet.

The workstation was successfully assembled by installing all hardware components in the correct sequence. After completing the assembly, the system powered on successfully, and all major components were detected, confirming proper installation and successful workstation assembly.

